PRODUCT DATA - FORMATTING AND VISUALIZATION

Project Objective:

The objective of this project is to perform data cleaning and preparation on a product dataset using Microsoft Excel. The dataset was analyzed to identify and handle missing values, standardize text formats, validate category information, review duplicate records, and improve overall data quality. Various Excel functions and tools were used to prepare the dataset for further analysis and reporting.

Column Name	Description Table
Product ID	A unique identifier assigned to each product. The Product ID contains information that was later separated into Manufacturing Date and Country Code during data transformation.
Product Name	The name of the product available in the dataset. This column was cleaned and standardized using text formatting functions.
Brand Name	The brand associated with each product. This column was later merged with Product Name to create a Product Brand field.
Quantity	Represents the number of units available or recorded for each product.
Price	Represents the price of each product. Missing values were identified and replaced using the average price of the column, and the data was formatted in currency format.
Category	Specifies the product classification, such as Electronics, Fashion, Kitchen, Accessories, and other categories. Missing values and inconsistencies in this column were corrected during data cleaning.

Data Cleaning and Transformation:

The dataset was examined for missing values, inconsistent text formats, duplicate records, and data quality issues. Missing values in the Price column were identified using ISBLANK() and replaced with the average value of the Price column. Missing Category values were identified using COUNTBLANK() and corrected using Find & Replace. Text inconsistencies were standardized using TRIM(), CLEAN(), and PROPER() functions.

Category values were validated using VLOOKUP() and corrected where necessary. Duplicate records were reviewed and retained after verification because they may represent valid repeated product transactions rather than data-entry errors. The Price column was formatted in Currency format, and the Manufacturing Date column was formatted in DD-MM-YYYY format. Conditional Formatting was applied using Data Bars and Color Scales in the Price column to improve visualization. A custom Conditional Formatting rule was also applied to highlight records where the Category is Electronics.

Tasks Performed:

1. Used **ISBLANK()** and **AVERAGE()** functions to identify and replace missing values in the Price column.

2. Used **COUNTBLANK()** and **Find & Replace** to identify and fill missing values in the Category column.

3. Used **TRIM()**, **CLEAN()**, and **PROPER()** functions to remove extra spaces, eliminate non-printable characters, and standardize text formatting.

4. Used **VLOOKUP()** and **Find & Replace** to validate category values and correct spelling mistakes or inconsistencies.

5. Reviewed duplicate records using Filter options. Duplicate rows were retained after verification because they may represent valid repeated product transactions.

6. Split the Product ID column into **Manufacturing Date** and **Country Code** using Excel data transformation techniques.

7. Merged the Brand Name and Product Name columns to create a new **Product Brand** column.

8. Applied **Currency Formatting** to the Price column and **DD-MM-YYYY Date Formatting** to the Manufacturing Date column.

9. Applied **Conditional Formatting** using Data Bars and Color Scales in the Price column to visualize price differences.

10. Applied a custom Conditional Formatting rule to highlight products belonging to the **Electronics** category.

Conclusion

The dataset was successfully cleaned and prepared using Microsoft Excel. Missing values were handled, text formats were standardized, category values were validated, and formatting techniques were applied. The final dataset is consistent, organized, and ready for further analysis.